

Deep Learning Lab Assignment - 1

The objective of this assignment is to investigate the geometric interpretation and learning behaviour of a Single Layer Perceptron through interactive visualization and experimental analysis.

Experiment I: Geometric Interpretation of a Single Layer Perceptron

1. Create a CSV file containing the complete truth table of a three-input AND gate. Load the dataset into the program and display it before proceeding with the remaining tasks.
2. Develop an interactive three-dimensional scatter plot using the three input features as the coordinate axes. Represent the two output classes using different colours or markers.
3. Provide interactive sliders to modify the three weights and the bias of the perceptron. Update the decision hyperplane in real time whenever any parameter is modified.
4. Investigate the effect of changing the weight vector and bias on the orientation and position of the decision hyperplane. Determine one set of perceptron parameters that correctly separates the dataset.

Experiment II: Learning Behaviour of a Single Layer Perceptron

1. Generate a two-dimensional linearly separable dataset consisting of 40 randomly generated samples, with 20 samples belonging to each class.
2. Display the generated dataset using a two-dimensional scatter plot.
3. Initialize the perceptron with randomly generated weights and bias, and implement the standard perceptron learning algorithm.
4. Animate the learning process by updating the decision boundary after every weight update while simultaneously displaying the dataset.
5. During training, display the current weight vector, bias, iteration number, decision boundary, and the number of misclassified samples. Highlight the training sample responsible for each weight update and display the angle between the current and previous weight vectors.
6. Continue the training process until all training samples are correctly classified or the maximum number of iterations is reached.
7. Plot the number of misclassified samples against the training iteration and discuss the convergence behaviour of the perceptron.
8. Repeat the experiment by varying the learning rate, random weight initialization, bias initialization, and feature scaling. Compare the convergence behaviour and explain the observed differences.

Algorithm: Perceptron Learning Algorithm

```
 $P \leftarrow \text{inputs with label } 1;$   
 $N \leftarrow \text{inputs with label } 0;$   
Initialize  $\mathbf{w}$  randomly;  
while !convergence do  
    Pick random  $\mathbf{x} \in P \cup N$  ;  
    if  $\mathbf{x} \in P$  and  $\sum_{i=0}^n w_i * x_i < 0$  then  
        |  $\mathbf{w} = \mathbf{w} + \mathbf{x}$  ;  
    end  
    if  $\mathbf{x} \in N$  and  $\sum_{i=0}^n w_i * x_i \geq 0$  then  
        |  $\mathbf{w} = \mathbf{w} - \mathbf{x}$  ;  
    end  
end  
//the algorithm converges when all the inputs  
are classified correctly
```
